

BIG DATA ANALYTICS

UNIT - I : Question Bank

(Short Questions | Long Answer Questions)

SECTION – A

SHORT ANSWER QUESTIONS (2–3 Marks Each)

Q1. What is Big Data? Give two examples.

Answer:

Big Data refers to extremely large datasets (Peta bytes and beyond – 10^{15} bytes) that cannot be stored or processed by traditional data management tools. These datasets come from diverse sources, grow rapidly, and require special frameworks to handle.

Examples:

- Facebook generates more than 500 terabytes of data every single day (photos, videos, messages, posts).
- The New York Stock Exchange generates approximately 1 terabyte of new trade data per day.

Q2. List and briefly explain the Four V's of Big Data.

Answer:

- Volume – The enormous amount of data generated (e.g., petabytes of social media data).
- Velocity – The speed at which data is created and needs to be processed in real-time (e.g., live stock prices).
- Variety – Data exists in multiple formats: structured (database tables), semi-structured (XML/JSON), and unstructured (images, videos).
- Veracity – The trustworthiness and quality/accuracy of data (e.g., filtering fake social media posts).

Q3. Differentiate between Structured, Semi-Structured, and Unstructured Data with examples.

Answer:

Type	Description	Example
Structured	Organized in fixed tabular format with defined schema. Stored in RDBMS.	Employee table, Excel sheet
Semi-Structured	No fixed schema but contains tags/metadata for partial organization. Called NoSQL data.	XML, JSON, Email, HTML
Unstructured	No defined format or schema. Also called 'dark data'. Very difficult to process.	Videos, images, audio, PDFs

Q4. What is Hadoop? What does HADOOP stand for?

Answer:

Hadoop stands for High Availability Distributed Object Oriented Platform. It is an open-source framework written in Java, designed for distributed storage and parallel processing of large datasets

across clusters of commodity (inexpensive) hardware. It was developed by Doug Cutting and Mike Cafarella and is now maintained by the Apache Software Foundation. It is used by companies like Facebook, Yahoo, Google, and Twitter to process petabytes of data cost-effectively.

Q5. What are the four types of Big Data Analytics? Give one example for each.

Answer:

- Descriptive Analytics – Answers 'What happened?' using historical data. Example: Google Analytics showing website traffic reports.
- Diagnostic Analytics – Answers 'Why did this happen?' using data mining. Example: An e-commerce site analyzing why sales dropped despite items in carts.
- Predictive Analytics – Answers 'What might happen?' using ML/AI. Example: Netflix predicting what shows a user will watch based on history.
- Prescriptive Analytics – Answers 'What should we do?' by recommending actions. Example: Self-driving cars analyzing the environment and deciding the next move.

Q6. Compare Hadoop with RDBMS on any four points.

Answer:

#	RDBMS	Hadoop
1	Handles structured data only	Handles structured, semi-structured, and unstructured data
2	Vertically scalable (expensive hardware)	Horizontally scalable (add commodity nodes)
3	Static schema required	Dynamic/schema-on-read
4	High cost (licensed software)	Free and open source

Q7. List the core components of Hadoop and their roles.

Answer:

- HDFS (Hadoop Distributed File System) – Storage unit. Stores large files by splitting them into 128 MB blocks across DataNodes.
- MapReduce – Processing unit. Processes data in parallel using Map() (filtering/sorting) and Reduce() (aggregation) phases.
- YARN (Yet Another Resource Negotiator) – Resource management unit. Allocates CPU/memory resources and schedules jobs across the cluster.
- Hadoop Common – Java libraries and utilities that support all other Hadoop modules.

Q8. What is HDFS? Name its two core components and their functions.

Answer:

HDFS (Hadoop Distributed File System) is the primary storage component of Hadoop. It is designed to store very large files across a cluster of commodity machines, splitting files into 128 MB blocks and replicating them for fault tolerance.

- NameNode (Master) – Stores metadata (file names, block locations, namespace). It does NOT store actual data. Only one active NameNode exists. It manages DataNodes and tracks block mapping.
- DataNode (Slave) – Stores the actual data blocks. Multiple DataNodes exist. They periodically send heartbeats and block reports to the NameNode to confirm their health and data inventory.

Q9. What is Rack Awareness in HDFS? Why is it important?

Answer:

Rack Awareness is the concept where the NameNode is aware of the physical rack location of each DataNode. A rack is a collection of ~40-50 DataNodes connected via the same network switch.

When storing replicas (default replication factor = 3), Rack Awareness ensures:

- 1st replica: Stored on the local rack.
- 2nd replica: Stored on another DataNode in the same rack.
- 3rd replica: Stored on a DataNode in a different rack.

Importance: If one rack fails (network switch down), data is still available on another rack, ensuring fault tolerance and high availability.

Q10. Write any five commonly used HDFS commands with their syntax and purpose.

Answer:

Command	Syntax / Example	Purpose
ls	<code>hdfs dfs -ls /</code>	Lists files and directories in HDFS
mkdir	<code>hdfs dfs -mkdir /newdir</code>	Creates a new directory in HDFS
copyFromLocal	<code>hdfs dfs -copyFromLocal /local/file /hdfs/dir</code>	Copies file from local to HDFS
copyToLocal	<code>hdfs dfs -copyToLocal /hdfs/file /local/dir</code>	Copies file from HDFS to local system
rm -r	<code>hdfs dfs -rm -r /newdir</code>	Recursively deletes a directory and contents

SECTION – B

LONG ANSWER QUESTIONS (10–15 Marks Each)

Q1. Explain the characteristics of Big Data (Four V's) in detail with suitable examples. Also discuss the importance of Big Data in various industries.

Part A: The Four V's of Big Data

Big Data is characterized by the following four dimensions:

1. Volume (Amount of Data)

Volume refers to the massive scale of data generated every second from countless sources. Traditional storage systems cannot handle this scale. Data in Big Data starts from Peta bytes (10^{15} bytes) and goes up to Yotta bytes (10^{24} bytes).

Example: Facebook generates over 500 TB of data daily. A single jet engine produces 10+ TB of data in just 30 minutes of flight. Google processes 99,000 searches every second.

2. Velocity (Speed of Data)

Velocity refers to the rate at which data is generated and needs to be processed in real-time. Sources like social media, IoT sensors, and financial markets produce data continuously at extremely high speeds.

Example: Twitter has ~500 million tweets per day. Stock exchanges require millisecond-level processing of transactions.

3. Variety (Different Types of Data)

Variety refers to the diverse formats in which data arrives — structured (database tables, Excel), semi-structured (XML, JSON, emails), and unstructured (images, videos, audio, PDFs). Around 80% of all digital data is unstructured.

Example: A hospital collects patient records (structured), lab reports as PDFs (unstructured), and HL7 messages (semi-structured) all at once.

4. Veracity (Quality/Trustworthiness of Data)

Veracity refers to the accuracy and reliability of data. Since data comes from multiple sources, it may contain errors, duplicates, or inconsistencies. Veracity ensures data is filtered and verified before analysis.

Example: Social media sentiment data may include fake accounts and bots. Veracity mechanisms filter out this unreliable data.

Part B: Importance of Big Data

- **Cost Savings:** Tools like Apache Hadoop and Spark store and process large volumes at low cost, replacing expensive proprietary systems.
- **Time Savings:** Real-time analytics allows instant decision-making. Example: Fraud alerts in banking.
- **Understanding Market Conditions:** Customer purchasing behavior analysis helps businesses forecast demand. Example: Amazon recommending products based on browsing history.
- **Social Media Listening:** Companies perform sentiment analysis to understand customer feedback. Example: Brand monitoring on Twitter.
- **Product Development:** Insights from usage data drive product innovation. Example: Netflix using viewer data to greenlight new shows.

Q2. Explain the Apache Hadoop Ecosystem in detail. Describe each component with its role and how they work together.

Introduction

The Apache Hadoop Ecosystem is a collection of open-source tools and frameworks that work together to extend Hadoop's capabilities across storage, processing, data access, and management.

1. HDFS – Hadoop Distributed File System (Data Storage)

HDFS is the core storage layer of Hadoop. It splits large files into 128 MB blocks and distributes them across DataNodes. The NameNode maintains metadata (block locations). HDFS provides high throughput access and fault tolerance via replication (default factor = 3).

2. HBase – NoSQL Database (Column Store)

HBase is a column-oriented, distributed NoSQL database that runs on top of HDFS. It is modeled after Google's BigTable and provides random, real-time read/write access to petabytes of structured data. Ideal for use cases like messaging systems and time-series data.

3. MapReduce – Data Processing

MapReduce is a parallel processing framework. The Map() function sorts and filters data into key-value pairs; the Reduce() function aggregates these pairs to produce the final result. Example: Counting word occurrences in millions of documents.

4. YARN – Yet Another Resource Negotiator

YARN manages cluster resources. It has three sub-components: Resource Manager (allocates resources globally), Node Manager (manages resources per node), and Application Master (negotiates resources for individual jobs). It acts as the OS for Hadoop.

5. Hive – SQL-like Query Interface

Hive is a data warehousing tool with HQL (HiveQL), a SQL-like language. HQL queries are automatically translated into MapReduce jobs. It is best for batch processing of large structured datasets. Used for ETL and data summarization.

6. Pig – Data Flow Scripting

Pig uses a scripting language called Pig Latin to process large datasets. It auto-generates MapReduce jobs. Pig is best for complex multi-step data transformations. Developed originally by Yahoo.

7. Sqoop – Data Transfer Tool

Sqoop (SQL-to-Hadoop) transfers bulk data between relational databases (MySQL, Oracle) and the Hadoop ecosystem (HDFS, Hive, HBase). Used for data import/export in ETL pipelines.

8. Flume – Data Ingestion Service

Flume collects, aggregates, and transfers large amounts of log/event data from multiple sources (web servers, social media) into HDFS. It is fault-tolerant and scalable.

9. ZooKeeper – Cluster Coordination

ZooKeeper manages and synchronizes distributed services across the Hadoop cluster. It provides configuration management, naming, and synchronization services, allowing developers to focus on application logic.

10. Oozie – Job Scheduling

Oozie is a workflow scheduler that manages and sequences Hadoop jobs. Multiple complex jobs (MapReduce, Hive, Pig, Sqoop) can be pipelined in a required order to achieve a desired outcome.

How They Work Together

Data flows into HDFS via Flume or Sqoop. YARN allocates processing resources. MapReduce, Hive, or Pig processes the data. Results are stored back in HDFS or HBase. Oozie schedules the entire workflow, while ZooKeeper ensures coordination.

Q3. Explain the Architecture and Design of HDFS in detail, including NameNode, DataNode, Secondary NameNode, Replication, and Rack Awareness.

Overview of HDFS Architecture

HDFS (Hadoop Distributed File System) follows a master-slave architecture. It is designed to store very large files (terabytes to petabytes) reliably across a cluster of commodity machines.

1. NameNode (Master)

The NameNode is the master server in HDFS. Key responsibilities:

- Manages the file system namespace (directory tree of files and folders).
- Stores metadata: file names, block IDs, block locations, replication factor.
- Maintains two files: FsImage (snapshot of namespace) and EditLogs (recent changes).
- Determines which DataNodes store which blocks.
- Does NOT store actual data — only metadata.

There is only ONE active NameNode per cluster (single point of coordination).

2. DataNode (Slave)

DataNodes are the worker nodes that store actual data blocks. Key responsibilities:

- Store and retrieve data blocks as instructed by the NameNode.
- Serve read and write requests from clients.
- Send periodic heartbeats (every 3 seconds) to the NameNode to signal health.
- Send block reports to let the NameNode know which blocks are stored.
- Can number from 1 to 500+ in large clusters.

3. Secondary NameNode

The Secondary NameNode is NOT a backup for the NameNode. It is a helper node that:

- Periodically downloads FsImage and EditLogs from the NameNode.
- Merges them locally to create an updated FsImage.
- Uploads the merged image back to the NameNode.

This prevents EditLogs from growing too large, reducing NameNode restart time.

4. Block Size and Storage

HDFS splits files into blocks of 128 MB by default (configurable up to 256 MB). Example: A 300 MB file is stored as three blocks — 128 MB + 128 MB + 44 MB — distributed across different DataNodes.

5. Replication

To ensure fault tolerance, each block is replicated across multiple DataNodes. Default replication factor = 3. If one DataNode fails, the block is still accessible from another.

Example: A 128 MB block with replication factor 3 uses 384 MB of total disk space across the cluster.

6. Rack Awareness

A Rack is a group of ~40-50 DataNodes connected by the same network switch. Rack Awareness is the NameNode's ability to place replicas intelligently:

- Replica 1: Local DataNode (same rack as the client).
- Replica 2: Different DataNode on the SAME rack.
- Replica 3: DataNode on a DIFFERENT rack.

This ensures data is available even if an entire rack's network switch fails, and reduces inter-rack network traffic.

Q4. Explain how a client reads and writes data in HDFS. Describe the step-by-step process for both READ and WRITE operations with diagrams description.

HDFS Read Operation

Step 1: The client calls the open() method on the DistributedFileSystem (DFS) object.

Step 2: The DFS object contacts the NameNode via RPC to get the locations (DataNode addresses) of the first few blocks of the requested file.

Step 3: The NameNode returns the block locations sorted by proximity to the client (rack awareness).

Step 4: An FSDataInputStream (wrapping DFSInputStream) is returned to the client.

Step 5: The client repeatedly calls read() on the DFSInputStream, which connects to the nearest DataNode and streams the block data.

Step 6: After reading the last byte of a block, DFSInputStream closes the connection and moves to the next block's DataNode.

Step 7: Once all blocks are read, the client calls close() on the stream.

HDFS Write Operation

Step 1: The client calls create() on the DistributedFileSystem, which sends an RPC request to the NameNode.

Step 2: The NameNode verifies that the file does not already exist and the client has permission. It creates a new file record. If the file exists or permissions are missing, an IOException is thrown.

Step 3: An FSDataOutputStream (wrapping DFSOutputStream) is returned to the client. The client calls write() to push data.

Step 4: DFSOutputStream breaks data into packets and places them in a DataQueue. A DataStreamer component dequeues packets and asks the NameNode to allocate new blocks, identifying suitable DataNodes for replication.

Step 5: A pipeline of DataNodes (equal to the replication factor, e.g., 3) is formed. The packet is written to the first DataNode, which forwards it to the second, which forwards it to the third.

Step 6: Each DataNode sends an acknowledgment packet back through the pipeline. These are tracked in the Ack Queue.

Step 7: Once all acknowledgments are received, the client calls close(), which flushes remaining packets and waits for final acknowledgment. The NameNode is then notified that the file write is complete.

Q5. Discuss the different use cases and applications of Big Data Analytics across industries. Also explain the brief history of Hadoop from 2002 to 2022.

Part A: Big Data Use Cases Across Industries

1. Banking, Financial Services & Insurance (BFSI)

BFSI is one of the most data-intensive sectors. Financial institutions use Big Data for:

- Fraud Detection: Analyzing millions of transactions in real-time to flag anomalies.
- Risk Management: Predicting creditworthiness and market risk.
- Customer Engagement: Personalizing offers based on spending behavior.

Example: Banks analyze ATM deposits, withdrawals, and online payments (KYC data) to detect suspicious activity instantly.

2. Healthcare

Healthcare generates data from patient records, lab reports, medical imaging, and insurance data. Big Data enables:

- Electronic Health Records (EHR): Central storage for patient history across hospitals.
- Improved Patient Predictions: Early identification of disease patterns.
- Real-Time Alerts: Monitoring ICU patients and alerting staff immediately.

3. Education

Educational institutions collect enrollment data, grades, and behavior patterns. Big Data helps:

- Create customized learning programs for individual students.
- Identify students at risk of dropping out early.
- Improve institutional performance through data-driven decision making.

4. Manufacturing

Manufacturers use sensor data from machines on production lines:

- Predictive Maintenance: Detecting equipment failures before they occur.
- Yield Improvement: Optimizing production processes using historical quality data.

- Supply Chain Management: Tracking raw materials and delivery schedules.

5. E-Commerce

E-commerce companies like Amazon and Flipkart use Big Data to predict customer trends, personalize recommendations, and optimize product pricing dynamically.

Part B: Brief History of Hadoop

Year	Milestone
2002	Doug Cutting & Mike Cafarella began the Apache Nutch web crawler project, facing scalability issues with big data.
2003	Google published its paper on the Google File System (GFS) describing distributed storage.
2004	Google released the MapReduce white paper describing parallel processing on large clusters.
2005	Doug Cutting introduced NDFS (Nutch Distributed File System) inspired by GFS + MapReduce.
2006	Doug Cutting joined Yahoo. Separated the distributed computing parts from Nutch to create Hadoop. Hadoop 0.1.0 released. Named after his son's toy elephant.
Jan 2008	Yahoo successfully tested Hadoop on a 1,000-node cluster.
Jul 2008	Yahoo released Hadoop as an open-source project to the Apache Software Foundation (ASF).
2009	Hadoop successfully sorted 1 Peta byte of data in less than 17 hours.
2011	Apache Software Foundation officially released Hadoop Version 1.0.
2012	Hadoop 2.0 released (introduced YARN for resource management).
2017	Hadoop 3.0 released with erasure coding and improved performance.
2022	Hadoop 3.2.4 released with stability improvements.